

Note 5.2 Sigma Notation and Limits of Finite Sums

Finite Sums and Sigma Notation

- Sigma notation:

$$\sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n.$$

in which $k = 1$ represents the index of summation k started at $k = 1$, n represents the index k ends at $k = n$, $\sum$ is the summation symbol (Greek letter sigma) and a_k is a formula for the k th term.

- Algebra Rules for Finite Sums:

- Sum Rule:

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$$

- Difference Rule:

$$\sum_{k=1}^n (a_k - b_k) = \sum_{k=1}^n a_k - \sum_{k=1}^n b_k$$

- Constant Multiple Rule:

$$\sum_{k=1}^n ca_k = c \cdot \sum_{k=1}^n a_k$$

for any number c .

- Constant Value Rule:

$$\sum_{k=1}^n c = n \cdot c$$

in which c is any constant value.

- Formula:

- the first n integers:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

- the first n squares:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

- the first n cubes:

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Limits of Finite Sums

- **Conclusion:** Any finite sum approximation is trapped between the lower and upper sum approximations. And define the area of region R as this limiting value.

Riemann Sums

- **Process:**
 - Suppose a bounded function f defined on a closed interval $[a, b]$, and then we choose $n - 1$ points $\{x_1, x_2, x_3, \dots, x_{n-1}\}$ on the interval $[a, b]$ and satisfying

$$a < x_1 < x_2 < \dots < x_{n-1} < b$$

to subdivide the interval $[a, b]$ into subintervals(not necessarily with the same width).

- To make the notation consistent, we denote a by x_0 and b by x_n such that

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$$

and the set

$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

is called the partition of $[a, b]$. And P divides $[a, b]$ into n subintervals

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n].$$

and we denote the $[x_{k-1}, x_k]$ as the k th subinterval of P .

- Then we denote the width of the k th subinterval as $\Delta x_k = x_k - x_{k-1}$ and then choose a point in the k th subinterval denoted by c_k .
- Then the area of the k th rectangle is $f(c_k) \cdot \Delta x_k$, whose sign depends on the $f(c_k)$.
- Finally we sum all the area of the rectangle to get

$$S_p = \sum_{k=1}^n f(c_k) \Delta x_k.$$

The sum S_p is called a **Riemann sum** for f on the interval $[a, b]$.

- **Norm:** The largest of all the subinterval widths of a partition P is called the norm of the partition P , written $\|P\|$.